1.

Machine Learning is a part of Artificial Intelligence where computers learn from data. It helps machines make decisions automatically without full programming.

There are mainly three types.

Supervised Learning uses labeled data. Example is prediction systems.

Unsupervised Learning uses unlabeled data and finds patterns.

Reinforcement Learning learns using rewards and punishment.

Machine Learning is used in many applications like recommendation systems and chatbots.

2.

Overfitting means the model learns too much from training data and gives poor results for new data.

Some ways to avoid overfitting are:

- More data
- Cross validation
- Dropout
- Regularization
- Early stopping

Overfitting reduces model performance. A balanced model is important for better accuracy.

3.

Reinforcement Learning is a Machine Learning method where the system learns by reward and punishment. The agent interacts with the environment and improves slowly.

The model uses trial and error learning. Good actions give rewards and bad actions give penalties.

Uses of Reinforcement Learning are:

- Self-driving cars
- Robotics
- Games
- Recommendation systems

For example, a robot learns walking after many attempts. Reinforcement Learning is useful in AI applications where continuous decision making is needed.

4.

Data preprocessing is very important in Machine Learning because raw data may contain missing values and errors.

Preprocessing includes:

- Cleaning data
- Removing duplicates
- Scaling values

The model does not behave the same with good and bad data.

Good data gives better prediction and accuracy. Bad data gives wrong output and poor performance.

For example, if incorrect data is given to the model, the result may also become incorrect. So preprocessing improves model quality.

5.

NLP stands for Natural Language Processing. It helps computers understand human language.

Applications are:

- Chatbots
- Translation
- Voice assistants

Embeddings are important because computers cannot understand words directly. Embeddings convert words into vectors.

Similar words have similar vectors. This helps the model understand meaning and relationships between words.

Word2Vec and BERT are examples of embeddings used in NLP systems.

6.

Machine Learning should be used when there is large data and prediction problems.

Uses:

- Recommendation systems
- Image recognition
- Fraud detection

Machine Learning should not be used for simple problems where normal programming is enough.

If there is less data, ML may not work properly. ML also requires training and maintenance.

So ML should be used only when necessary and useful for the problem.